1 Mathematical Framework

1.1 Model (Prototype)

The model depends on hyperparameters

$$\sigma_1, \dots, \sigma_K, \lambda_1^{(0)}, \dots, \lambda_K^{(0)}, \lambda_1^{(1)}, \dots, \lambda_K^{(1)}, n_1, \dots, n_K, c_1, \dots, c_K$$

$$1, 2, \dots, K \text{ arms}, \ 1, 2, \dots, T \text{ Time Horizon}$$

$$Y_{i,t}^{(\sigma_i)} \sim N(Z_{i,t}, \sigma_i^2) \text{ Reward of arm } k \text{ at time } t$$

$$Z_{i,t} = \mathbb{E}(Y_{i,t}^{(\sigma)}) \in \mathcal{Z}$$

$$\mathcal{Z} = \left\{ Z \in \{0,1\}^{K \times T} : \sum_{t=1}^{T} Z_{i,t} = n_i, \ \frac{1}{T} \sum_{t=1}^{T-1} Z_{i,t} Z_{i,t+1} \geq c_i \right\}$$

$$a_t \subseteq \{1, \dots, K\} \text{ active arms at time } t$$

$$\text{Let } 1_{i,t} = 1(i \in a_t), \text{ Let } H_i^{(0)}(t) = \min\{s \geq 1 : 1_{i,t+s} \neq 1_{i,t}, \ 1_{i,t} = 0\} \text{ Sleeping holding time}$$

$$\text{Let } H_i^{(1)}(t) = \min\{s \geq 1 : 1_{i,t+s} \neq 1_{i,t}, \ 1_{i,t} = 1\} \text{ Awake holding time}$$

$$H_i^{(j)}(t) \sim \text{Geom}(\lambda_i^{(j)}), \ i.e \ \mathbb{P}(H_i^{(j)}(t) = s) = (1 - \lambda_i^{(j)})^{s-1} \lambda_i^{(j)}, \ \ s \geq 1$$

$$\text{Let } X_t \text{ be the decisions of the algorithm } f \in \mathcal{F}_t \ s.t \ \mathcal{F}_t = \left\{ f : \{Y_{X_s,s}^{(\sigma_{X_s})}, X_s\}_{s=1}^{t} \to a_t \right\} \ \forall \, t \in \{1, \dots, T\}$$

We now formalize the regret:

$$R_T = \mathbb{E}\left(\sum_{t=1}^{T} Y_{i^*,t}^{(\sigma_{i^*})} 1_{i^*,t} - \sum_{t=1}^{T} Y_{X_t,t}^{(\sigma_{X_t})} \right) = \sum_{t=1}^{T} Z_{i^*,t} 1_{i^*,t} - \sum_{t=1}^{T} Z_{X_t,t}$$

$$i^* = \arg\max_{i \in [K]} \sum_{t=1}^{T} Z_{i,t} 1_{i,t}$$

2 Supplementary Material for Code

- `src/hybrid.py <class object> gauss_hybrid_model` : Refer to A1

3 A1

The class depends on three subsystems:

- `src/hybrid.py <function object> gauss_hybrid_model/gen_A`
- `src/hybrid.py <function object> gauss_hybrid_model/gen_Z`
- `src/hybrid.py <function object> gauss_hybrid_model/gen_Y`

3.1 Generating A (two state markov chain)

$$A_{i,t} = \{1, 0\}^{K \times T}$$

$$\text{Initialise } A_{i,1} = 1 \ \forall \ i$$

$$\text{For } t = 2, 3, \dots, T$$

$$U_{i,t} \sim Unif(0, 1)$$

$$A_{i,t} = 1(A_{i,t-1} = 1 \cap U_{i,t} \geq \lambda_i^{(1)}) + 1(A_{i,t-1} = 0 \cap U < \lambda_i^{(0)})$$

3.2 Generating Z (partitions + stars and bars selection)

$$M \sim Unif(1, \ldots, M_{max}),\ M_{max} = \min(n_i - c_i T, T - n_i + 1)$$

Choose $M - 1$ dividers uniformly without replacement from $\{1, \ldots, n_i - 1\}$

:

$$d_1 < d_2 < \cdots < d_{M-1}, \quad d_k \sim \text{Unif}\binom{[n_i - 1]}{M - 1}$$

Setting $d_0 = 0$ and $d_M = n_i$, the cluster sizes are:

$$s_j = d_j - d_{j-1}, \quad j = 1, \ldots, M$$

This samples uniformly over all compositions of n_i into M positive parts:

$$s_1 + s_2 + \cdots + s_M = n_i, \quad s_j \geq 1$$

with each of the $\binom{n_i - 1}{M - 1}$ compositions equally likely.

3.3 Generating Y

$$Y_{i,t}^{(\sigma_i)} \sim N(Z_{i,t}, \sigma_i)$$

Note: in the programming language, we denote an inactive arm as NaN and passing a NaN through a gaussian rng doesn't change the value.